

A complete mapping witness for the conjecture as written

CONJECTURE 00000000127 | TRUE AS WRITTEN

Submitted by SucRunBug - AI-assisted mathematical work and formalization

Conjecture

For every prime $p > 3$, an exponent d outside $\{1, p-2\}$ makes both $x \mapsto x^d$ and $x \mapsto x^d + x$ permutations of $\mathbb{F}_p$.

Proof as written

Take $d=p$. This differs from 1 and $p-2$. Fermat's identity gives $x^p=x$ for every field element, including zero. Thus the two maps are the identity and multiplication by 2. Both are bijective, since 2 is nonzero for $p > 3$.

Interpretation boundary

The file does not impose $d < p$ or a canonical exponent range modulo $p-1$. The exponent p is equivalent as a polynomial function to exponent 1, so this proof does not produce a nontrivial new complete mapping. If the intended question excludes all exponents congruent to 1 modulo $p-1$, or restricts d to a canonical range, this witness is excluded and that modified conjecture is not settled here.

Formalization scope

Lean uses the standard field $\mathbb{Z} \bmod p$, quantifies over every prime $p > 3$, proves the two literal exclusions for $d=p$ and proves bijectivity of both maps. The interpretation limitation is part of this submission.

Reproduction

Pinned Lean and Mathlib: v4.33.0. In lean4/: lake exe cache get; lake build; lake env lean Check.lean. The independent reproduce.py checks finite instances; the written general proof and Lean theorems settle the infinite statements.

Source: repository conjectures/00000000127.md; main branch df87a69ed4814eeb34ad161879b53db10c44fdc6.